

NAN PHYU SIN MAUNG

Data Engineer

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PROFESSIONAL SUMMARY

A fresh Computer Science graduate specializing in Big Data, with strong foundations in Python, SQL, Apache Spark, Hadoop, AWS, and Azure. Experienced in data pipelines, big data processing, and cloud platforms through academic projects. Eager to bring technical skills and analytical thinking to a data engineering role.

EDUCATION

Diploma of Information Technology
Singapore Institute of Management
2023–2024

Bachelor of Computer Science (Big Data)
University of Wollongong
2024 – 2026

SKILLS

Programming Languages

- Python, Java, C++

Data Visualization

- Microsoft Excel, Matplotlib, Seaborn

Other Technologies

- HTML, CSS, JavaScript, Machine Learning, Git/GitHub, ETL Pipelines

Databases & Big Data

- SQL, MySQL, PostgreSQL, MongoDB, Apache Hadoop, Apache Spark

Cloud Platforms

- Microsoft Azure, Amazon Web Services (AWS)

PROJECTS

Tasking — Smart Task Allocation Web App (Final Year Project)

Next.js, TypeScript, Supabase (PostgreSQL), Vercel, Stripe, OpenAI, Playwright, Vitest

- Built full-stack SME workforce management platform supporting 8 user roles across recruitment, scheduling, task assignment, attendance, and reporting
- Integrated OpenAI for AI-assisted job posting, task assignment, and anomaly detection in reports
- Enforced Row Level Security on all database tables with comprehensive E2E, performance, security, and scalability testing

Emochi — AI Emotion Support App (24-hr Hackathon | Microsoft Stack Award & Community Choice Award)

Next.js, Azure AI Foundry, SQL Server, Prisma, Tailwind CSS, Microsoft Entra ID

- Built full-stack multi-agent AI app featuring 8 AI emotion characters, a live debate arena with real-time streaming, and Microsoft Entra ID authentication
- Won 2 awards out of all hackathon participants within 24 hours

Multilingual Neural Machine Translation — Comparative Study

Python, HuggingFace Transformers, MarianMT, mBART-50, NLLB-200, Google Colab

- Compared 3 NMT models fine-tuned across 6 translation directions on 100M sentence pairs, achieving BLEU scores up to 40.12
- Built memory-efficient training pipeline using Seq2SeqTrainer, Adafactor optimizer, and FP16

Hope Spring — Online Fundraising Platform

TypeScript

- Built full-stack fundraising platform with role-based access for 3 user types, campaign management, donation tracking, and automated daily/weekly/monthly reporting

Taxi-v3 Reinforcement Learning

Python, OpenAI Gymnasium, NumPy, Matplotlib

- Trained Q-learning agent across 30,000 episodes to optimize passenger routing in a 500-state environment with learning curve visualization

Oops Too Slow — Reaction Game (24-hr Hackathon)

React, Vite, JavaScript, MediaPipe, Firebase

- Built full-stack multiplayer reaction game with keyboard, mouse, and hand gesture input using MediaPipe, with Firebase-powered live leaderboard

Crimson Midnight — AI Murder Mystery Game

Next.js, JavaScript, CSS

- Developed front-end for AI-powered detective game with dynamic NPC dialogue, multiple story scenarios, and decision-based endings; deployed on Vercel